

Vinay Shanker

🌐 Vinay Shanker | ✉️ vinaymuddhe@gmail.com | 🔗 Vinay Shanker | 📄 Leetcode | 🏠 Github |

PROFESSIONAL SUMMARY

Proactive beginner Full Stack Developer with hands-on experience in Spring Boot and React ecosystems, actively building and deploying robust applications. Contributed to a patent-pending AI platform and developed a personal portfolio, demonstrating proficiency in RESTful API design and PostgreSQL.

EDUCATION

Jain (Deemed-to-be University), Faculty of Engineering & Technology <i>B.Tech in Information Science & Engineering</i>	Aug 2023 – Present Bengaluru, Karnataka
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RELEVANT COURSEWORK

- Data Structures & Algorithms • Database Management Systems • Web Technologies • Operating Systems
- Object-Oriented Programming • Computer Networks • Software Engineering • Artificial Intelligence

PROJECTS

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| Solo Tutor — AI-Powered Learning Platform <i>Python, FastAPI, React, Supabase</i> | 2024 – Present |
| <ul style="list-style-type: none">• Patent Filing In Progress · 5 Student Co-Inventors• Built a multi-modal RAG platform enabling hallucination-free Q&A from documents and YouTube URLs using persistent vector embeddings (pgvector) in Supabase.• Engineered automated quiz generation and YouTube transcript analysis with Whisper-based audio fallback; named co-inventor on the patent application at Jain University. | |
| Distributed Rate Limiter & API Gateway <i>Java, Spring WebFlux, Redis, Docker</i> | 2026 |
| <ul style="list-style-type: none">• Built a high-throughput API Gateway using Token Bucket rate-limiting with Spring WebFlux for non-blocking I/O.• Used Redis with Lua scripts for atomic distributed state management; load-tested with JMeter. | |
| Event-Driven E-Commerce Order Engine <i>Spring Boot, Kafka, PostgreSQL, Docker</i> | 2026 |
| <ul style="list-style-type: none">• Designed a fault-tolerant microservices architecture separating Order, Inventory, and Payment domains using the Saga Pattern.• Orchestrated asynchronous event streams with Apache Kafka; implemented distributed tracing with Zipkin. | |

TECHNICAL SKILLS

Languages: Java, JavaScript, HTML, CSS, SQL
Frameworks & Libraries: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST APIs, JWT Authentication, React.js
Databases: PostgreSQL, MongoDB
AI: Spring AI (learning), RAG, pgvector
Developer Tools: Git, GitHub, VS Code, Maven, Gradle, Docker

ACHIEVEMENTS & EXTRACURRICULAR

- Named co-inventor on the Solo Tutor Invention Disclosure Form and patent application filed at Jain University — recognising contribution to the AI-powered quiz generation module.
- Self-learning Spring Boot and Spring AI ecosystem through project-based practice, building REST APIs and exploring AI integration for backend services.